

Achromatopsia Clinical Trials

ClinicalTrials.gov Summary | Query: ACHROMATOPSIA OR CNGB3 | 13 studies | Retrieved June 2026

1. Active / Ongoing Gene Therapy Trials

NCT02599922 — AAV Gene Therapy for CNGB3 Achromatopsia (AGTC-401)

Sponsor: Beacon Therapeutics

Status: Active, Not Recruiting

Phase: Phase 1/2

Started: April 2016

Gene target: CNGB3 — accounts for ~50% of achromatopsia cases

Intervention: Subretinal injection of AGTC-401 (AAV-CNGB3), single eye

Design: Non-randomised, open-label. Primary endpoint: safety. Secondary: efficacy.

Link: clinicaltrials.gov/study/NCT02599922

NCT02935517 — AAV Gene Therapy for CNGA3 Achromatopsia (AGTC-402)

Sponsor: Beacon Therapeutics

Status: Active, Not Recruiting

Phase: Phase 1/2

Started: August 2017

Gene target: CNGA3 — second most common cause of achromatopsia (~25% of cases)

Intervention: Subretinal injection of AGTC-402 (AAV-CNGA3), single eye

Design: Non-randomised, open-label. Primary endpoint: safety. Secondary: efficacy.

Link: clinicaltrials.gov/study/NCT02935517

NCT02610582 — rAAV.hCNGA3 Gene Therapy — Bilateral Subretinal Delivery

Sponsor: STZ eyetrial (Germany)

Status: Active, Not Recruiting

Phase: Phase 1/2

Started: November 2015

Gene target: CNGA3

Intervention: Single bilateral subretinal injection of rAAV.hCNGA3

Population: Adults and paediatric patients

Link: clinicaltrials.gov/study/NCT02610582

Note: One of the few trials to explore bilateral subretinal delivery.

2. Completed Gene Therapy Trials

NCT03001310 — Gene Therapy for Achromatopsia (CNGB3) — MeiraGTx

Sponsor: MeiraGTx UK II Ltd

Status: Completed

Phase: Phase 1/2

Started: January 2017

Gene target: CNGB3

Intervention: AAV-CNGB3 retinal gene therapy, subretinal injection

Link: clinicaltrials.gov/study/NCT03001310

NCT03758404 — Gene Therapy for Achromatopsia (CNGA3) — MeiraGTx

Sponsor: MeiraGTx UK II Ltd

Status: Completed

Phase: Phase 1/2

Started: August 2019

Gene target: CNGA3

Intervention: AAV-CNGA3 retinal gene therapy, subretinal injection

Link: clinicaltrials.gov/study/NCT03758404

NCT01648452 — CNTF Implants for CNGB3 Achromatopsia

Sponsor: National Eye Institute (NEI)

Status: Completed

Phase: Phase 1/2

Started: July 2012

Gene target: CNGB3 (neuroprotection approach, not gene replacement)

Intervention: Intraocular CNTF sustained-release implant, surgically placed in one eye

Population: Adults ≥ 18 with confirmed CNGB3 mutations, BCVA $\leq 20/100$

Link: clinicaltrials.gov/study/NCT01648452

Note: Tests whether CNTF (ciliary neurotrophic factor) can stabilise or improve cone function — a pharmacological/protein-based approach distinct from gene replacement.

3. Non-Gene Therapy / Pharmacological Trials

NCT04041232 — Glycerol Phenylbutyrate (PBA) for ATF6-mediated Achromatopsia

Sponsor: Columbia University

Status: Suspended

Phase: Early Phase 1

Started: April 2025

Gene target: ATF6 — a rarer subtype (~2% of achromatopsia cases)

Intervention: Oral glycerol phenylbutyrate (PBA), 3x daily, 4.5–11.2 mL/m²/day

Mechanism: ATF6 mutations cause ER stress in photoreceptors; PBA acts as a chemical chaperone to reduce ER stress and potentially restore retinal function

Monitoring: Minimum 3 clinic visits with retinal function tests, fundus exam, and imaging

Link: clinicaltrials.gov/study/NCT04041232

Note: Drug repurposing approach — PBA is already FDA-approved for urea cycle disorders. Distinct from the AAV gene therapy programme. Currently suspended.

4. Terminated / Discontinued

NCT03278873 — Long-Term Follow-Up Gene Therapy Study (CNGB3 and CNGA3) — MeiraGTx

Sponsor: MeiraGTx UK II Ltd

Status: Terminated

Started: June 2017

Purpose: Long-term follow-up of patients from MeiraGTx's CNGA3 and CNGB3 AAV gene therapy trials

Link: clinicaltrials.gov/study/NCT03278873

5. Observational / Natural History / Registry Studies

NCT04124185 — Natural History Study for Achromatopsia — MeiraGTx

Sponsor: MeiraGTx UK II Ltd

Status: Completed

Started: September 2011

Purpose: Pre-clinical phenotypic characterisation to identify suitable trial candidates. Advanced multimodal imaging to probe structure-function relationships.

Relevance: Foundational work that directly fed into MeiraGTx's Phase 1/2 interventional trials.

Link: clinicaltrials.gov/study/NCT04124185

NCT01846052 — Clinical and Genetic Characterisation of CNGB3 Achromatopsia

Sponsor: Beacon Therapeutics

Status: Completed

Started: June 2013

Purpose: Identified and phenotyped CNGB3 achromatopsia patients every 6 months for up to 1.5 years

Relevance: Screening / natural history study that preceded Beacon's Phase 1/2 AGTC-401 trial.

Link: clinicaltrials.gov/study/NCT01846052

NCT07085533 — Natural History Study of Inherited Retinal Diseases

Sponsor: Zhongmou Therapeutics

Status: Recruiting

Started: July 2025

Purpose: ~200 subjects with varied IRDs + controls. Examines chromatic vision deficits vs. structural retinal biomarkers (ellipsoid zone integrity on SD-OCT). Validates the Moji Low-Vision Color Discrimination Test.

Relevance: Includes achromatopsia within IRD cohort; generates structural-functional correlates useful for outcome measures in future trials.

Link: clinicaltrials.gov/study/NCT07085533

NCT02435940 — Inherited Retinal Degenerative Disease Registry (My Retina Tracker)

Sponsor: Foundation Fighting Blindness

Status: Recruiting (ongoing since June 2014)

Purpose: Patient-initiated registry capturing genotypic/phenotypic data, genetic results, family history, and clinical measurements across a broad range of IRDs including achromatopsia.

Relevance: Accelerates clinical trial recruitment and establishes genotype-phenotype correlations. Open-access resource for investigators.

Link: clinicaltrials.gov/study/NCT02435940

NCT06491615 — eyeGENE Stage 3 — Expansion of DNA and Data Repository

Sponsor: National Eye Institute (NEI)

Status: Recruiting

Started: July 2024

Purpose: Expanding the NIH eyeGENE biobank with targeted accrual for rare inherited eye diseases. Collects saliva/blood DNA samples and clinical imaging data.

Diseases covered: Aniridia, Best disease, blue-cone monochromacy, corneal dystrophies, albinism, achromatopsia

Relevance: Long-running NIH infrastructure; data feeds into trial recruitment and genotype-phenotype research.

Link: clinicaltrials.gov/study/NCT06491615

6. Summary Table

NCT ID	Short Title	Gene	Approach	Status	Phase	Sponsor
NCT02599922	AAV CNGB3 (AGTC-401)	CNGB3	AAV subretinal	Active, not recruiting	Ph 1/2	Beacon Therapeutics
NCT02935517	AAV CNGA3 (AGTC-402)	CNGA3	AAV subretinal	Active, not recruiting	Ph 1/2	Beacon Therapeutics
NCT02610582	rAAV.hCNGA3 bilateral	CNGA3	AAV subretinal bilateral	Active, not recruiting	Ph 1/2	STZ eyetrial
NCT03001310	AAV CNGB3 — MeiraGTx	CNGB3	AAV subretinal	Completed	Ph 1/2	MeiraGTx
NCT03758404	AAV CNGA3 — MeiraGTx	CNGA3	AAV subretinal	Completed	Ph 1/2	MeiraGTx
NCT01648452	CNTF implant CNGB3	CNGB3	Neuroprotection	Completed	Ph 1/2	NEI
NCT04041232	PBA for ATF6	ATF6	Drug repurposing	Suspended	Early Ph 1	Columbia University
NCT03278873	LT follow-up CNGB3/CNGA3	Both	Follow-up	Terminated	—	MeiraGTx
NCT04124185	Natural history	—	Observational	Completed	—	MeiraGTx
NCT01846052	Clinical/genetic char.	CNGB3	Observational	Completed	—	Beacon Therapeutics
NCT07085533	IRD natural history	—	Observational	Recruiting	—	Zhongmou
NCT02435940	My Retina Tracker	—	Registry	Recruiting	—	Fdn Fighting Blindness
NCT06491615	eyeGENE Stage 3	—	Biobank/registry	Recruiting	—	NEI

7. Key Themes for Dissertation

- Gene targets:** CNGB3 (~50% of cases) and CNGA3 (~25%) dominate the trial landscape. ATF6 (~2%) has one early pharmacological trial.
- Delivery method:** All gene therapy trials use AAV (adeno-associated virus) vectors delivered by subretinal injection — the standard surgical approach for retinal gene therapy.
- Trial phases:** All interventional trials are Phase 1/2, reflecting the still-early clinical stage of achromatopsia gene therapy.
- Main sponsors:** Two industry leaders — Beacon Therapeutics (CNGA3 + CNGB3) and MeiraGTx (CNGA3 + CNGB3) — with academic/NIH work on complementary approaches (CNTF, PBA, registries).
- Alternative approach:** The Columbia PBA trial is notable as a non-GT approach targeting ER stress in a rarer ATF6 subtype — currently suspended.
- Observational infrastructure:** Multiple completed and active natural history/registry studies (MeiraGTx, Beacon, NEI eyeGENE, Foundation Fighting Blindness) provided phenotypic data that informed and continues to inform trial design and recruitment.